A brief overlook at the effects of patents in pharmaceutics

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Introduction

Innovation doesn't come out of nowhere. It’s a constant process made by people and requires resources like time, equipment and capital that could also be spent elsewhere. In a competitive market new actors will join in on a market until all the profits are depleted. Naturally this creates a barrier to innovating; Why spend resources trying to invent something, if everyone will immediately copy your findings and you end up at the starting point again? Sure, the humanity might be better off for it, but once everyone knows something, nobody will have an edge over each other. Of course it will take some time for others to catch up, but in a world without laws for protection you must work under the assumption that any lead you can get via innovation won’t last too long.

The most widely adopted solution to incentivize inventions are patents, which date back in some form all the way to ancient times. The general idea is that if you invent something, you can patent it. The patent in turn gives you a right to exclude others from using the invention (= a monopoly right) for the patent's length, after which it becomes a public good that anyone can use. As such, all patents always come with a trade-off. On one hand you are giving incentives for innovation, but on the other hand you are creating temporary monopolies, which come with the problems monopolies have.

However. especially in recent years the traditional idea that patents incentivize innovation has been challenged more and more. As more studies have been made, the answer whether this actually holds up has become less and less clear. Most economists agree with that when implemented correctly, patents can be a net positive. The current state of patents however is not as clear, and especially in recent years the developments have been worrying. In this project I’m going to focus on the effects patents currently have on pharmaceutics and explore whether we could reach globally better outcomes by shaking things up.

The effects of patents in pharmaceutics

In pharmaceutics, the cost of innovation is much higher that on most other fields. According to a 2016 study, the success rate of experiments is only four percent. Thus, every successful experiment needs to cover the cost of itself and approximately 24 failed experiments before turning a profit. While the reported production costs vary a lot between studies, recent estimates have found the mean costs of bringing a new medicine to market to be anywhere between $310 and $2800 million and a 2020 paper found the median cost to be $1.1 billion.

It goes without saying that these are huge sums, and the pharmaceutical companies often argue that they couldn't be justified if not for the incentives created by the monopoly right you gain via patents. However most pharmaceutical companies seem to be quite profitable under current regulations with companies like Johnson & Johnson raking in profit in tens of billions annually.

One might ask, what is the problem in all of this? JNJ have clearly pushed the medical field forward and are now simply reaping the benefits of their innovations. If this was any other field, we would be celebrating them!

To give an answer will require us to delve into ethics, which is a deep, deep field with very shaky foundations. However, generally speaking most countries value human lives, and maybe more specifically their citizens above all else. After the second the world war UN made the universal human rights declaration that have been almost universally signed, which among many others declare medical care as a universal human right (Article 25).

Now if we take this as our starting point, it doesn't take long to notice how patents in pharmaceutics contradict this article. By giving monopoly rights over sections of healthcare, the monopolists have complete power over things that could be lifesaving to the customer. This can result in people without sufficient wealth or safety nets to be left without treatment and potentially lose their lives. Like everywhere else, the supply and demand curves play an effect here as well. A monopolist has complete control over supply and as we've learned, this doesn't ever result to efficient outcomes for customers. In this context, the effects aren't merely some numbers on a paper, but actual human lives.

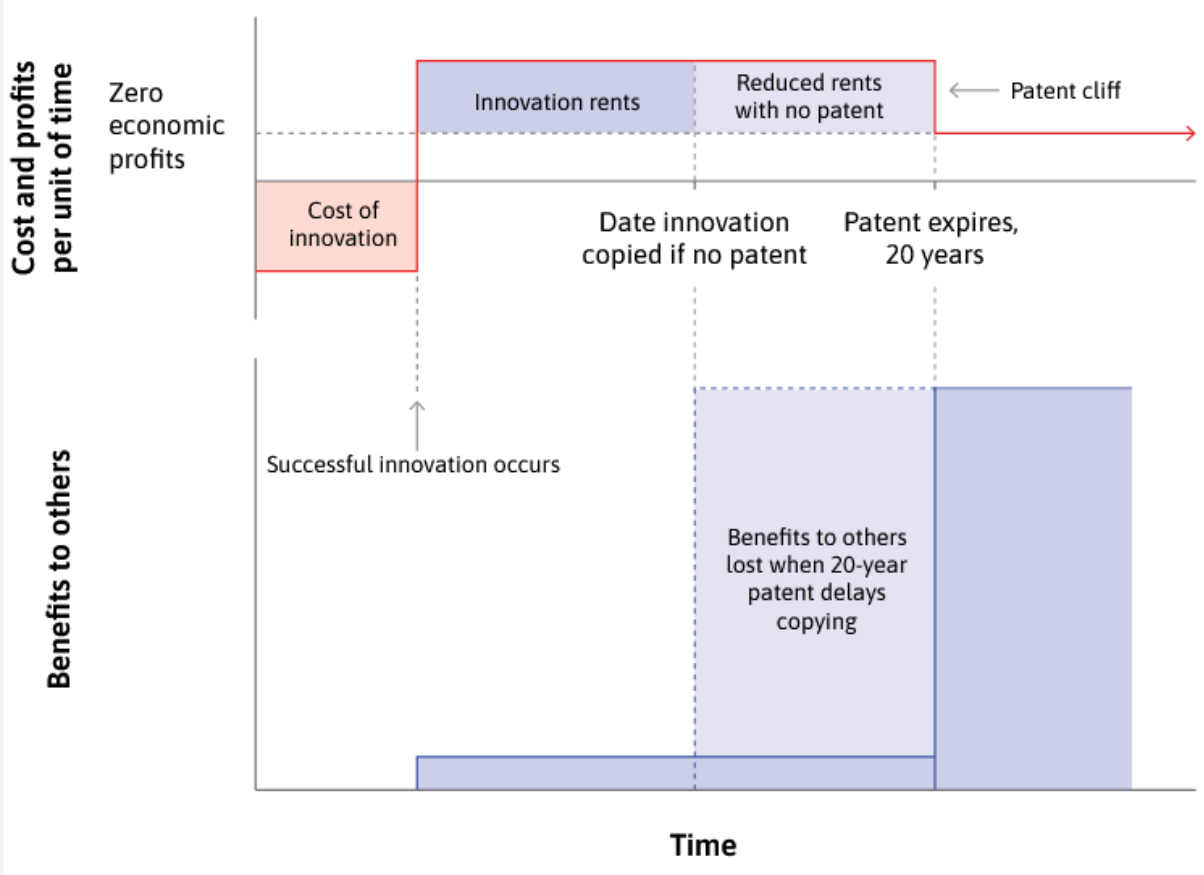
So, are there any solutions to this problem? On one hand our goal should be to make sure that as many people as possible receive the drugs that will improve their lives. At the same time, we want to maximize the rate of innovation so that we also receive more supplemental treatments in the future. With patents you are always balancing between these two goals, but as we'll see, there might be alternative routes we haven't explored yet that could lead to better results. If the huge profits that pharmaceutic companies

rake haven't given it up already, it should be safe to say that we currently aren't in the most efficient position.

Possible improvements

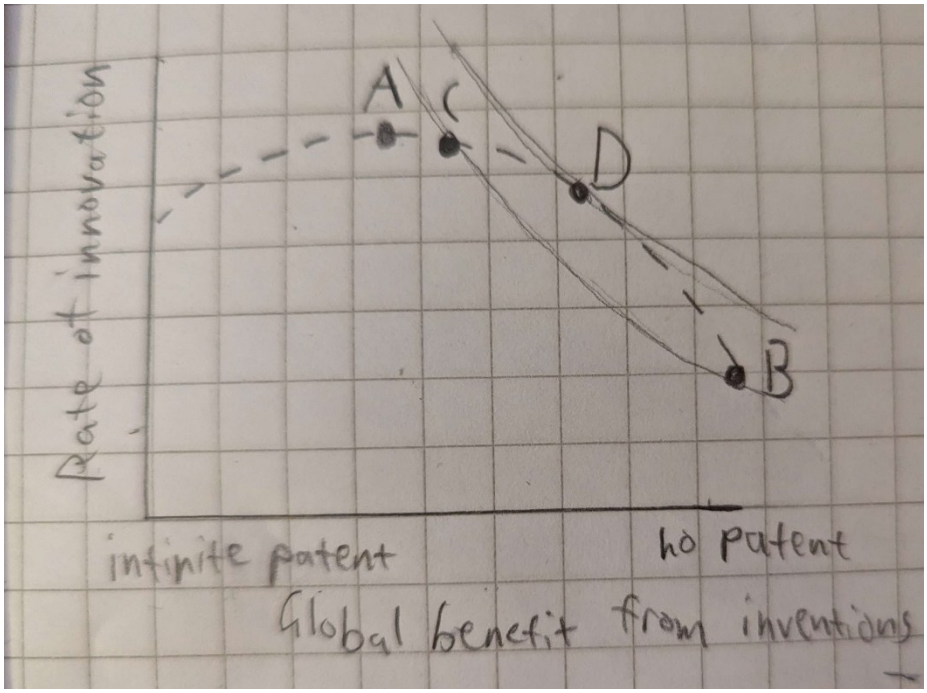
The basics of patents and innovation apply in pharmaceutics and to start things off, it should be stated that patents are not a necessity for innovation. Even without patents the innovator would still gain the innovation rents that last until the invention gets copied elsewhere. In the case of pharmaceutics, the first mover advantage was estimated to be around four years according in a study by Jean O. Lanjouw (1998). In simpler terms this means that it would take approximately four years for competing products to get to the shelves of pharmacies, which already creates some incentives to innovate.

The graph below shows the basic idea behind all this; An invention will cost resources, but once a breakthrough gets made the inventor will receive innovation rents until eventually the invention gets copied by others and profits get depleted in competition (Though society will be better off as showcased in the lower half of the graph since we get more efficiency).



Patents could be seen as an artificial extension to the innovation rents. They increase the gain that the inventor gets but decrease the overall social gain of the invention by keeping others from using it. F.M. Scherer, a professor at Harvard University argues that patents are particularly important when it comes to pharmaceutics due to the high innovation costs, but is that really the case? Could we find a better balance between the incentives to innovate and social gains than patents?

A comfortable first option would be to examine the power of patents and whether we could get to better outcomes without giving up on the system. As we established before, medicine as it stands is a highly profitable business. If we imagine that the trade-off between innovation rate and global benefits of the innovation follow a curved line, we might be able to get to a higher indifference curve by trading some of the innovation rate for a lot of global benefits. In other words, we may stand to gain more than we would lose by weakening patents as illustrated in the graph below.



- Point A: Highest possible rate of innovation
- Point B: Highest possible global benefit from inventions (no patents)
- Point C: Hypothetical point that showcases current patent strength
- Point D: The point that reaches the highest indifference curve

In principle, this would be very hard to implement. Boldrin and Levine (2013) point out that patents are a technical subject that very few of the consumers know about. As such the players that are influencing patents mainly consist of people who find it in their interest to have patents around in the first place. This has resulted in patents becoming stronger over time, due to a skewed equilibrium point, that isn't representative of what's best to the population as a whole. Reversing this progress would be no small feat since

you'd have to go against very powerful players. Regardless, this would in principle be the easiest way to improve the situation. Weaker, and perhaps more selective monopolies would probably have a much greater positive effect in accessibility than a negative effect in hindering innovation.

Another option would be to go against the views of Scherer and abolish patents entirely. The graph above shows that this probably isn't a great idea, but it doesn't account in that we could introduce alternative systems in place of patents that create incentives with possibly fewer downsides.

Economists like Kremer and Williams (2009), among others have argued that a system that's built around prices could very well be better than the current system that's built around timed monopolies. The basic idea being that you offer money for anyone who comes up with a solution to the problem you're having. In the context of pharmaceutics this would probably mean setting up a price for a solution to some disease or mutation or setting up a fund to reward medical innovations more broadly. As long as the prices are adequate, it should create incentives and since the findings will become public, we'll end up with competition sooner that with patents (Recall the four-year estimation from above, whereas drug patents last for 20 years).

It could even make sense for these prices to be greater than the current gain pharmaceutical companies get from patents. A study by Chaudhuri, Goldberg, and Gia (2006) found that the introduction of pharmaceutic patents in just one drug family in India resulted in a welfare loss of $300 million with the private gain being only $20 million. Basic game theory should tell us that a pareto improvement is possible, whereby paying any sum between 20 and 300 million to the patent owners would result in both parties being better off.

It could also be possible to combine the two. For instance, we could declare that treatments deemed extremely important are public goods and reward finding those via prices, while offering patents to less important innovations.

Conclusion

Based on my relatively limited research in these matters and my shallow overall grasp on economics, I cannot in good faith give confident proposals or claim to be an expert by any stretch of the word. Regardless of the individual aspects I highlighted in this project, the broader point I want to make is that it would be foolish to view our current system that revolves around monopolies as one that couldn't be improved in a number of ways. Especially in the context of pharmaceutics and medicine, lives are in a very literal sense on the line. As such we should make every effort to get into socially better outcomes.

Sources

The Economy V1 by CORE-ECON with an emphasis on unit 21

Lecture slides from this and the past courses

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Kallis elämä (HS 30.1.2024, Anni Lassila)

Estimated Research and Development Investment Needed to Bring a New Medicine to Market, 2009-2018 (JAMA Network, Olivier J. Wouters, Martin McKee and Jeroen Luyten)

The Introduction of Pharmaceutical Product Patents in India: "Heartless Exploitation of the Poor and Suffering"? (nber, Jean O. Lanjouw)

Johnson & Johnson 2023 Results (JNJ)

Universal Declaration of Human Rights (UN)

No AI tools were used in the making of this project.